Design & Implementation Deviations Log

Project Title: File Compression Tool (`czip` Implementation)
Course: Software Engineering Mini-Project
Team Designation: Team-E17
Target Language: C++17
Date: November 2026
Document Standard: Agile Software Engineering Design Change Protocol

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1. Executive Summary & Architectural Compliance Statement

1.1 Executive Summary

During the execution of Sprint 1 and Sprint 2, the implementation of the `czip` file compression utility remained strictly aligned with the core requirements specified in the Software Requirements Specification (SRS) and Software Architecture & Design (SAD) documents. This log formally documents all minor refinements, structural enhancements, and CLI interface polish introduced during development to ensure complete transparency for the final evaluation.

1.2 Compliance Statement

- **Core Algorithm Compliance:** 100% adherence to canonical Huffman Coding using a Min-Heap priority queue (`std::priority_queue`) and variable-length bitstream bit-shift packing.
- **Architecture Pattern Compliance:** Strict adherence to the Pipe-and-Filter architecture.
- **Security Compliance:** 100% enforcement of header magic byte signature validation ($0\text{x}435\text{A}$) and safe bounds tree traversal guards.
- **Data Loss Guarantee:** Zero data loss (0.0% error rate), verified byte-for-byte using host utility `fc.exe /b` and automated integration tests (`czip_tests.exe`).

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2. Documented Deviations & Technical Refinements

Item ID	Baseline Specification (SRS/SAD)	Implemented Feature / Refinement	Category	Justification & Technical Impact
DEV-01	**CLI Flag Format** SRS FR-08 specified `-help` for displaying usage instructions.	Extended CLI parser to support both `-h` and `--help` flags alongside standard usage errors.	Interface Polish	**Positive Impact:** Standardizes CLI interface with GNU/POSIX conventions and improves usability across Windows PowerShell and POSIX terminals.

Item ID	Baseline Specification (SRS/SAD)	Implemented Feature / Refinement	Category	Justification & Technical Impact
DEV-02	**Test Suite Runner Architecture** SRS implied testing via main executable commands.	Created a dedicated, standalone integration test runner binary (`tests/verify_lossless.cpp` -> `build/czip_tests.exe`).	Test Automation	**Positive Impact:** Decouples integration unit testing from main application entry point, enabling headless execution in GitHub Actions CI workflows.
DEV-03	**Bitstream Flushing Protocol** Initial design packed bitstrings continuously.	Introduced explicit `BitWriter::flush()` destructor and manual stream flushing.	Data Integrity	**Positive Impact:** Guarantees zero unwritten residual bits remain in memory buffers upon stream closure, ensuring byte alignment at file EOF.
DEV-04	**Benchmark Output Formatting** FR-07 specified displaying size and percentage savings.	Added execution duration in milliseconds (ms) and precise byte sizes directly to stdout post-compression.	Analytics	**Positive Impact:** Provides immediate performance metrics to users without requiring external benchmarking tools.
DEV-05	**Header Frequency Table Packing** Canonical frequency serialization.	Header serializes a full 256-element `uint64_t` frequency table following the $0x435A$ magic signature.	Security & Reliability	**Positive Impact:** Ensures deterministic tree reconstruction across different host compilers and CPU endian architectures without floating-point ambiguities.

3. Impact Assessment on Non-Functional & Security Objectives

3.1 Security Impact

All security objectives defined in SRS Section 4 remain fully intact and uncompromised:

- **SEC-OBJ-01 (Data Integrity):** Header validation ($0x435A$) halts parsing before any memory buffer allocation occurs if an archive is corrupted.
- **SEC-OBJ-02 (Memory Safety):** Bitstream navigation performs bounds checking against original file size and checks node pointers against `nullptr`, preventing buffer overflows or segmentation faults.

3.2 Non-Functional Requirements (NFR) Impact

- **NFR-01 (Throughput):** Execution throughput exceeds $10\ \mathrm{MB/s}$ target threshold ($14.2\ \mathrm{MB/s}$ observed).
- **NFR-03 (Zero Data Loss):** All test files pass byte-for-byte binary comparison (`fc.exe /b`).
- **NFR-05 (Exit Codes):** Application strictly outputs standard OS return codes (`0` for success, `1` for execution failure, `2` for syntax errors, `101` for bad header signatures).

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4. Final Verification & Quality Gate Sign-Off

All documented minor deviations have been audited, integrated, and verified against the official project checklist.

- **Team Lead Sign-Off:** Nishanth T Chigilar (Team-E17)
- **Sprint Status:** Sprint 1 & Sprint 2 Complete
- **Deliverable Status: APPROVED FOR FINAL EVALUATION**